

SHAGUN PATEL

San Diego, California

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Summary

Computer Science graduate student specializing in AI-driven software systems, with a strong focus on agentic workflows and LLM-powered applications.

Education

San Diego State University

Master of Science in Computer Science (GPA:3.91)

Aug 2025 – Present (Expected May 2027)

San Diego, CA

Lok Jagruti University

Bachelor of Engineering in Computer Science (CGPA: 9.27)

Sep 2021 – May 2025

Ahmedabad, India

Skills

- **Programming:** Python, C, C++, Java, JavaScript, HTML/CSS
- **Backend & APIs:** FastAPI, Django, REST APIs, Microservices
- **Databases:** MySQL, MongoDB
- **Cloud & DevOps:** Linux (Ubuntu), AWS (EC2, S3 – basic), Git, CI/CD fundamentals
- **Tools:** GitHub, Excel, Word, PowerPoint, Google Workspace
- **AI/ML:** TensorFlow, PyTorch, Numpy, Scikit-learn, NLP, Deep Learning, Computer Vision, LLM Integration
- **AI Agents:** Claude, GitHub Copilot, Cursor, ChatGPT

Experience

San Diego State University

Sep 2025 – Present

Research Assistant RTEC Lab - Real-Time Embedded and Control Systems Laboratory

San Diego, CA

- Developed and tested autonomous vehicle systems using **ROS2 and Gazebo**, focusing on real-time perception and control
- Integrated **LiDAR and camera sensor data** for environment understanding and navigation in GPS-denied settings
- Implemented **sensor fusion and decision-making frameworks** for reliable real-time autonomous operation

Karmaleen Technology

Jun 2024 – Mar 2025

Software Engineer

Ahmedabad, Gujarat

- Designed and deployed Python microservices with FastAPI on Ubuntu servers, **reducing API response latency by 35%**
- Built and integrated NLP-powered chat agents using ChatGPT APIs, **lowering client response time from 2 minutes to less than 30 seconds**.
- Automated workflows with MySQL optimizations that **cut query execution time by 40%**, saving nearly 10 hrs/week in manual effort.

Karmaleen Technology

Dec 2023 – May 2024

Software Engineer Intern

Ahmedabad, Gujarat

- Built ML models (TensorFlow, Scikit-learn) and deployed prototypes on **Ubuntu environment**.
- Implemented MySQL indexing and query optimization, **cutting data retrieval from 12s to 5s**.
- Developed RESTful APIs with FastAPI/Django and optimized backend logic.

Projects

COVID-19 X-ray Image Generator 🌀 | *Python, TensorFlow, Keras, GANs*

- Built a **GAN to generate synthetic X-ray images** (COVID-19, Lung Opacity, Normal, Viral Pneumonia) to improve dataset diversity for medical analysis.

Deepfake Detection – Real vs. Fake Classification 🌀 | *TensorFlow, Keras, OpenCV, NumPy*

- Developed a **Convolutional Neural Network (CNN)** model attaining **92% accuracy** in distinguishing real from manipulated images, validated through rigorous testing, and identified the root cause of model performance bottleneck.

Bitcoin Price Prediction | *Python, LSTM, Sklearn, TensorFlow*

- Implemented an **LSTM network** to forecast Bitcoin prices with visualizations for **trend analysis and decision-making**.